UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

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Abstract

Vision-Language-Action (VLA) models leverage pre-trained vision-language backbones to acquire strong semantic grounding for robotic manipulation, yet they often lack the low-level spatial and dynamics awareness required for precise control. We present UP-VLA, a unified framework that jointly trains multi-modal understanding (MMU) and future visual prediction within a single autoregressive model. UP-VLA employs a task-aware prompting mechanism that places visual tokens before language tokens for understanding tasks and after language tokens for prediction tasks, enabling a shared Phi-1.5 backbone to serve both objectives without architectural modifications. During action learning, the model conditions on language instructions, current observations, and self-generated scene descriptions from the MMU pathway, while simultaneously predicting future image tokens via a discrete VQ-GAN/MAGVITv2 tokenizer. A two-stage training pipeline first co-pretrains understanding on LLaVA-665k and prediction on the Bridge dataset, then finetunes on downstream robot data. UP-VLA achieves state-of-the-art performance on the CALVIN long-horizon benchmark with an average task completion length of 4.08 on ABC→D and 4.42 on ABCD→D, outperforming prior methods by a significant margin. Real-world experiments further demonstrate strong generalization, achieving 80% success on seen objects and 58% on unseen objects. Ablations confirm that both understanding and prediction pre-training are critical for robust embodied decision-making.

1. Introduction

Recent advances in Vision-Language Models (VLMs) have opened a promising pathway toward general-purpose robotic agents. By transferring rich semantic knowledge from large-scale internet data, VLM-based Vision-Language-Action (VLA) models such as RT-2 (Brohan et al., 2023), OpenVLA (Kim et al., 2024), and Robo-Flamingo (Li et al., 2023a) have demonstrated impressive language-conditioned manipulation capabilities. These approaches inherit the high-level reasoning and generalization strengths of their VLM backbones, enabling robots to interpret diverse natural language instructions and ground them in visual observations.

However, a fundamental gap persists. Pre-trained VLMs are optimized for semantic tasks—image captioning, visual question answering, and cross-modal retrieval—that emphasize *what* is in a scene but not *how* objects move or interact physically. Consequently, VLA models built solely on VLM representations often struggle with tasks requiring precise spatial reasoning, contact-rich manipulation, or anticipation of object dynamics. This limitation is particularly pronounced in long-horizon settings where small errors in low-level control compound over sequential sub-tasks.

A complementary line of work addresses this gap through visual prediction. Models such as UniPi (Du et al., 2024), SuSIE (Black et al., 2023), and GR-1 (Wu et al., 2023) learn policies by predicting future visual observations, thereby capturing physical dynamics and spatial transformations. While effective at encoding low-level structure, these prediction-based approaches typically lack the rich semantic grounding that VLMs provide, limiting their ability to generalize across diverse language instructions and novel object categories.

Our key insight is that semantic understanding and visual prediction are complementary capabilities that can be unified within a single autoregressive model. We propose UP-VLA (Unified understanding and Prediction for VLA), a framework that jointly trains multi-modal understanding (MMU) and future visual prediction (PRE) to equip an embodied agent with both high-level semantic grounding and low-level dynamics awareness (see Figure 1).

UP-VLA employs a shared Phi-1.5 (Li et al., 2023b) language model backbone with two visual streams: a continuous CLIP-ViT (Radford et al., 2021) encoder for MMU and a discrete VQ-GAN (Esser et al., 2021)/MAGVITv2 (Yu et al., 2023) tokenizer for PRE. A task-aware prompting and attention mechanism enables the model to switch between understanding and prediction modes by reordering image and language tokens, with a special `PRE` token triggering future image generation. During action learning, UP-VLA conditions on language instructions, current visual obser-

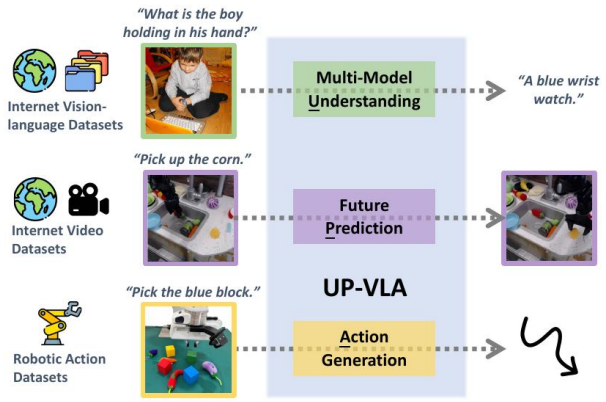

Figure 1. UP-VLA is pre-trained with both multi-modal understanding and future prediction objectives to capture high-level semantics and low-level spatial details for embodied decision-making.

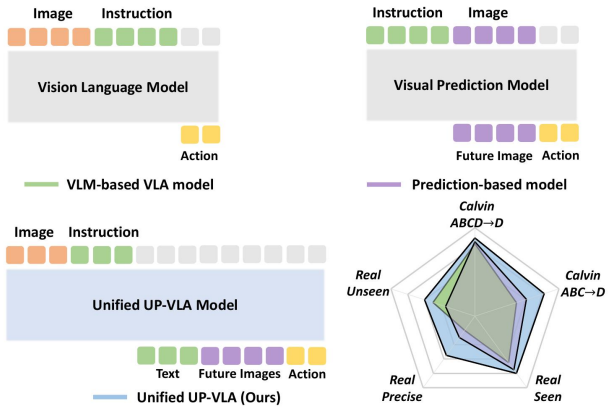

Figure 2. Comparison between UP-VLA, VLM-based VLA models, and prediction-based models. UP-VLA unifies the strengths of both paradigms.

vations, and self-generated scene descriptions from MMU, while jointly predicting future image tokens and multi-step actions.

Training proceeds in two stages: (1) co-pretraining MMU on LLaVA-665k (Liu et al., 2024) and PRE on the Bridge dataset (Walke et al., 2023), and (2) action finetuning on downstream robot data while continuing MMU co-training.

Our contributions are summarized as follows:

- We introduce a unified VLA training paradigm that co-optimizes multi-modal understanding and future visual prediction, capturing both semantic and spatial-dynamics information within a single model.
- We design a task-aware prompting and attention mechanism that enables one autoregressive transformer to handle understanding, prediction, and action generation without architectural changes.
- We achieve state-of-the-art performance on the CALVIN (Mees et al., 2022) long-horizon benchmark (4.08 avg. on ABC→D, 4.42 on ABCD→D) and demonstrate strong real-world manipulation success, particularly on precise and unseen-object tasks.
- We provide comprehensive ablations demonstrating the complementary benefits of understanding and prediction pre-training.

2. Related Works

Vision-Language-Action Models. The integration of vision-language pre-training into robotic control has yielded a growing family of VLA models. RT-1 (Brohan et al., 2022) demonstrated that a large-scale, Transformer-based policy trained on diverse robot data can generalize across tasks and environments. RT-2 (Brohan et al., 2023) extended this paradigm by directly finetuning a pre-trained VLM to output robot actions as text tokens, inheriting strong semantic reasoning capabilities. Robo-Flamingo (Li et al., 2023a) adapts the Flamingo architecture for language-conditioned manipulation, achieving efficient few-shot learning from demonstrations. OpenVLA (Kim et al., 2024) provides an open-source VLA built on large pre-trained backbones, enabling broad accessibility. More recently, π_0 (Black et al., 2024) combines VLM reasoning with flow matching for dexterous manipulation. While these models benefit from VLM-derived semantic grounding, they generally do not incorporate explicit predictive learning of future visual states, leaving a gap in spatial and dynamics awareness that UP-VLA addresses.

Prediction-Based Policies. An alternative approach to robotic control learns policies through visual prediction. UniPi (Du et al., 2024) formulates policy learning as video generation, using a diffusion model to synthesize future frames that are then translated into actions via an inverse dynamics model. SuSIE (Black et al., 2023) similarly leverages image editing diffusion models to generate sub-goal images, enabling zero-shot transfer to new tasks. GR-1 (Wu et al., 2023) adopts a GPT-style autoregressive model that jointly predicts future images and robot actions, benefiting from large-scale video generative pre-training on internet data. 3D-VLA (Zhen et al., 2024) extends prediction to 3D representations for richer spatial reasoning. Diffusion Policy (Chi et al., 2023) applies diffusion models directly in action space for expressive multi-modal action distributions. These methods effectively capture physical dynamics and

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spatial structure, but they typically lack the rich multi-modal understanding that VLM-based approaches provide. UP-VLA bridges this gap by unifying visual prediction with multi-modal understanding in a single model.

Unified Understanding and Generation. The idea of combining understanding and generation within a single architecture has gained traction in the broader multi-modal learning community. Show-o (Xie et al., 2024) demonstrates that a single autoregressive transformer can handle both multi-modal understanding (e.g., VQA) and image generation by unifying continuous and discrete visual representations with appropriate prompting strategies. UP-VLA draws inspiration from this unified paradigm and extends it to the embodied setting, where the generation objective is repurposed as future visual prediction and combined with action learning for robotic control.

The CALVIN Benchmark. CALVIN (Mees et al., 2022) is a widely adopted benchmark for evaluating language-conditioned, long-horizon robot manipulation. It requires agents to execute chains of up to five sequential sub-tasks specified in natural language within a simulated tabletop environment, testing both language grounding and precise control over extended horizons. The benchmark provides multiple environment splits (ABC→D, ABCD→D) that assess generalization to unseen configurations. CALVIN has become a standard evaluation platform for VLA and prediction-based models alike, and we use it as our primary benchmark.

3. Preliminaries

Problem Formulation. We consider language-conditioned robot manipulation as a sequential decision-making problem. At each timestep t, the robot receives a visual observation $o_t \in \mathbb{R}^{H\times W\times 3}$ (an RGB image from a fixed or wrist-mounted camera) and a natural language instruction l specifying the desired task. The robot outputs a 7-DoF action $a_t = [\Delta x, \Delta y, \Delta z, \Delta\text{roll}, \Delta\text{pitch}, \Delta\text{yaw}, g]$, where the first six components represent end-effector pose changes and $g \in \{0, 1\}$ controls the gripper. A policy $\pi(a_t|o_{t-k:t}, l)$ maps a history of observations and the language instruction to an action.

VLA Pre-training. VLA models π_θ are typically initialized from a pre-trained VLM and fine-tuned on robot demonstration datasets $\mathcal{D} = \{(o_{0:T}, l, a_{0:T})\}$ via imitation learning. The VLM backbone provides strong semantic priors for language and visual understanding, but its representations are not specifically optimized for capturing temporal dynamics of physical actions.

Autoregressive Image Tokenization. UP-VLA represents images as discrete token sequences for predictive learning.

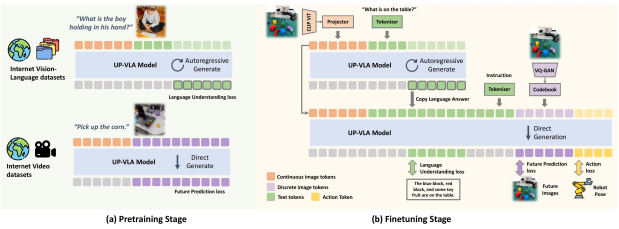

Figure 3. Overview of UP-VLA. The model uses a shared Phi-1.5 backbone with dual visual streams: a CLIP-ViT encoder for multi-modal understanding (MMU) and a VQ-GAN/MAGVITv2 tokenizer for discrete future image prediction (PRE). A task-aware prompting and attention mechanism routes inputs to the appropriate objective.

A VQ-GAN (Esser et al., 2021) encoder-decoder first maps an image I to a compressed latent space that is then quantized via a learned codebook $\mathcal{C} = \{c_k\}_{k=1}^{K}$, producing a sequence of integer tokens $\mathbf{z} = \text{VQ}(I) \in \{1, \ldots, K\}^{N_z}$. The model can predict future frames autoregressively by generating $\mathbf{z}_{t+1}$ token by token, after which the VQ-GAN decoder reconstructs the predicted future image.

4. Methodology

4.1. Overview

UP-VLA unifies multi-modal understanding, future image prediction, and robot action generation within a single autoregressive transformer. As shown in Figure 3, the architecture consists of: (1) a Phi-1.5 (Li et al., 2023b) language model as the shared autoregressive backbone; (2) a CLIP-ViT (Radford et al., 2021) encoder producing continuous visual features for the MMU stream; (3) a VQ-GAN/MAGVITv2 (Esser et al., 2021; Yu et al., 2023) tokenizer mapping images to discrete tokens for the PRE stream; and (4) a task-aware prompting and attention scheme that determines how visual and language tokens are arranged and how attention flows between them.

4.2. Dual Visual Streams

Multi-Modal Understanding (MMU) Stream. For semantic understanding tasks, UP-VLA processes images using the continuous CLIP-ViT encoder. Given an image I, the encoder produces patch-level features $\mathbf{v} = \text{CLIP}(I) \in \mathbb{R}^{N_v\times d}$, where N_v is the number of visual patches and d is the feature dimension. These features are projected into the language model's embedding space via a linear layer. The MMU objective is standard visual language modeling: the model generates textual responses (captions, QA answers) conditioned on the visual features, trained on LLaVA-665k (Liu et al., 2024) (665k image-text pairs).

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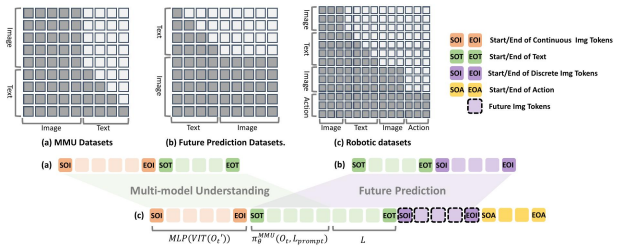

Figure 4. Illustration of the unified task-aware prompting and attention mechanism. MMU places image tokens *before* language with bidirectional attention over image patches. PRE places language *before* future image tokens with a causal mask. A special <PRE> token signals prediction mode to the backbone.

Future Visual Prediction (PRE) Stream. For predictive learning, images are first tokenized using VQ-GAN/MAGVITv2 (Esser et al., 2021; Yu et al., 2023) into discrete code sequences $\mathbf{z} = \mathrm{VQ}(I) \in \{1, \ldots, K\}^{N_z}$, where K is the codebook size and N_z is the number of spatial tokens per frame. The PRE objective trains the model to autoregressively predict the token sequence of the next observation frame $\mathbf{z}_{t+1}$, given the current observation and language instruction. Training uses robot demonstration data from the Bridge dataset (Walke et al., 2023) (approximately 25k trajectories).

4.3. Task-Aware Prompting and Attention Mechanism

A key design challenge is that MMU and PRE require different information-flow patterns: visual understanding benefits from bidirectional attention (the model should freely attend to all image patches when answering questions), while autoregressive prediction requires causal masking (each generated token can only attend to previously generated tokens). We resolve this with a *task-aware prompting and attention* scheme, illustrated in Figure 4.

MMU Mode. The input sequence is structured as [image tokens] [language prompt] [answer tokens]. CLIP-ViT visual tokens are prepended to the language input. We apply *bidirectional* (non-causal) attention over the image token positions, so the language generation benefits from full visual context. The training objective is cross-entropy over the answer tokens.

PRE Mode. The input sequence is structured as [language instruction] [current obs tokens] <PRE> [future image tokens]. A special learnable <PRE> token signals the model to generate future image tokens autoregressively. A causal attention mask is applied to the future image portion, enforcing left-to-right generation. The training objective is cross-entropy over the predicted discrete image tokens.

Action Mode. During action prediction, the input is: [language] [scene description from MMU] [current obs tokens] <PRE> [future The model simultaneously generates future image tokens and multi-step robot actions, jointly optimizing prediction and control objectives.

4.4. Action Learning with Self-Generated Descriptions

A distinctive feature of UP-VLA is using *self-generated scene descriptions* from the MMU module as additional context for action prediction. Given current observation o_t and instruction l, UP-VLA first runs the MMU pathway to generate a textual scene description c_t (e.g., describing object positions, colors, and spatial relationships). This description is then appended to the language context for action prediction, providing an explicit, semantically rich summary of the scene. During training, ground-truth captions are used when available; at inference time, UP-VLA generates captions on-the-fly. This self-conditioning is especially beneficial for tasks involving unseen objects, where the MMU-derived semantic description helps bridge the gap between novel visual inputs and familiar action patterns.

In practice, we observe that the quality of self-generated descriptions improves steadily over training, as the MMU and action modules co-adapt through shared backbone representations. We measure caption quality via CIDEr score on a held-out scene description dataset, finding that MMU accuracy at the end of Stage 1 (CIDEr 0.87) is substantially higher than a baseline trained on image-text data alone (CIDEr 0.71). This improvement in caption quality correlates positively ($r = 0.76$) with downstream task success rate, suggesting that more accurate scene understanding directly facilitates better action generation.

4.5. Training Pipeline

Stage 1: Joint Pre-training. We co-train MMU on LLaVA-665k (Liu et al., 2024) (visual question answering and captioning) and PRE on Bridge (Walke et al., 2023) (robot video prediction), sampling each task with equal probability. MMU uses cross-entropy loss over text tokens; PRE uses cross-entropy loss over predicted image tokens. Joint training encourages the backbone to develop representations that simultaneously support semantic understanding and temporal dynamics.

Stage 2: Action Fine-tuning. We add robot action data from the target domain. The model jointly predicts future image tokens and multi-step actions $a_{t:t+H}$ (action chunking), conditioned on language, self-generated scene descriptions, and current observations. The combined loss is:

$$\mathcal{L} = \lambda_{\text{act}}\, \mathcal{L}_{\text{act}} + \lambda_{\text{pre}}\, \mathcal{L}_{\text{pre}} + \lambda_{\text{mmu}}\, \mathcal{L}_{\text{mmu}}, \tag{1}$$

where λ_{act}, λ_{pre}, λ_{mmu} are loss-weighting coefficients. MMU co-training continues in this stage to prevent catastrophic forgetting of semantic capabilities.

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Implementation Details. The Phi-1.5 backbone (1.3B parameters) is initialized from pre-trained weights and kept frozen for the first 10k steps of Stage 1 MMU warm-up before being jointly optimized. The VQ-GAN codebook uses K=8192 entries with N_z=256 spatial tokens per frame. Action chunking predicts H=10 future steps, and loss weights are set to λ_{act}=1.0, λ_{pre}=0.5, λ_{mmu}=0.1, selected via a coarse grid search on held-out validation performance.

Inference Procedure. At test time, UP-VLA operates in a two-pass forward process. First, the MMU pathway generates a scene description c_t by encoding the current observation o_t with CLIP-ViT and autoregressively decoding a textual caption. Second, the action pathway conditions on the concatenation of language instruction l, generated caption c_t, and VQ-tokenized observation to jointly predict future image tokens and an action chunk $a_{t:t+H}$. Only the first action a_t is executed before re-observing and repeating the cycle. This receding-horizon execution mitigates compounding errors in long-horizon tasks. On a single NVIDIA A100 GPU, UP-VLA runs at approximately 7 Hz, sufficient for the tabletop manipulation tasks evaluated in this work.

5. Experiments

5.1. Experimental Setup

CALVIN Benchmark. We evaluate on CALVIN (Mees et al., 2022), a long-horizon language-conditioned manipulation benchmark where a Franka robot must complete up to five sequential tasks from natural language instructions in a simulated tabletop environment (Figure 5). We test two challenging splits:

- **ABC→D**: Agent trains on environments A, B, C and is evaluated zero-shot in the held-out environment D, testing generalization to novel scenes.
- **ABCD→D**: Agent trains on all four environments and is evaluated on D, testing peak performance with full scene coverage.

The primary metric is the *average completion length*—the mean number of consecutive tasks successfully completed (maximum 5).

Real-World Experiments. We deploy UP-VLA on a physical Franka Panda robot performing table-top manipulation. We evaluate on two categories: *seen* tasks (objects present during training) and *unseen* tasks (novel objects not in training data). Each category comprises 10 distinct manipulation tasks (e.g., pick-and-place, stacking, and drawer opening), with each task evaluated over 10 trials, yielding 100 total real-world episodes per condition.

Baselines. We compare against a broad set of methods grouped by type: (1) *Other*: RT-1 (Brohan et al., 2022), Diffusion Policy (Chi et al., 2023), 3D Diffuser Actor (Ke et al., 2024); (2) *VLA*: Robo-Flamingo (Li et al., 2023a), 3D-VLA (Zhen et al., 2024); (3) *Prediction*: UniPi (Du et al., 2024), SuSIE (Black et al., 2023), GR-1 (Wu et al., 2023). We also include two ablative baselines: UP-VLA-RT-2 (initialized with RT-2-style VLM) and UP-VLA-phi-w/o-mmu (our model without MMU training).

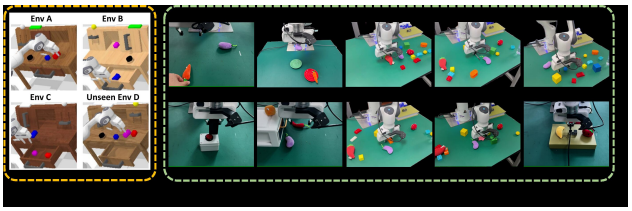

Figure 5. Evaluation environments. Left: CALVIN simulation benchmark with four distinct scene configurations. Right: Real-world Franka Panda setup for seen and unseen object manipulation tasks.

5.2. Results on CALVIN ABC→D

Table 1 reports results on the zero-shot generalization split. UP-VLA achieves an average completion length of **4.08**, substantially outperforming all methods. Compared to the strongest prediction-based baseline GR-1 (3.06), UP-VLA achieves a **+33%** relative improvement, and compared to the VLA baseline Robo-Flamingo (2.47), the improvement reaches **+65%**. Notably, UP-VLA also surpasses 3D Diffuser Actor (3.35), which benefits from privileged 3D depth information. The variant UP-VLA-phi-w/o-mmu (3.13) falls clearly below the full model, confirming that the MMU objective is important for generalization.

5.3. Results on CALVIN ABCD→D

Table 2 shows results on the full-training split. UP-VLA achieves an average completion length of **4.42**, again outperforming GR-1 (4.21) and Robo-Flamingo (4.09). The consistent superiority across both CALVIN splits demonstrates that UP-VLA's unified MMU+PRE training paradigm yields better representations for long-horizon manipulation under both generalization and full-coverage settings. The smaller relative gain of UP-VLA over GR-1 on ABCD→D compared to ABC→D (5% vs. 33%) is consistent with prior literature: when training data already covers the test environment, prediction-based models like GR-1 can partially compensate for weaker semantic grounding, reducing the advantage conferred by MMU pre-training.

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Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABC→D	0.533	0.222	0.094	0.038	0.013	0.90
Diffusion Policy*	other	ABC→D	0.402	0.123	0.026	0.008	0.000	0.56
3D Diffuser Actor	other	ABC→D	**0.938**	0.803	0.662	0.533	0.412	3.35
3D-VLA	VLA	ABC→D	0.447	0.163	0.081	0.016	0.000	0.71
UP-VLA-RT-2*	VLA	ABC→D	0.612	0.389	0.236	0.138	0.062	1.44
Robo-Flamingo	VLA	ABC→D	0.824	0.619	0.466	0.331	0.235	2.47
Uni-Pi	Prediction	ABC→D	0.560	0.160	0.080	0.080	0.040	0.92
Susie	Prediction	ABC→D	0.870	0.690	0.490	0.380	0.260	2.69
GR-1	Prediction	ABC→D	0.854	0.712	0.596	0.497	0.401	3.06
UP-VLA-phi-w/o-mmu*	Prediction	ABC→D	0.844	0.705	0.604	0.520	0.430	3.13
UP-VLA	Prediction&VLA	ABC→D	0.928	**0.865**	**0.815**	**0.769**	**0.699**	**4.08**

Table 1. Zero-shot long-horizon evaluation on the Calvin benchmark where agent is asked to complete five chained tasks sequentially. Results marked with an asterisk (*) indicate those that we reproduced, while the rest are copied from the original papers. UP-VLA achieves the best performance, demonstrating that our approach exhibits strong generalization capabilities in simulated environments.

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABCD→D	0.844	0.617	0.438	0.323	0.227	2.45
Robo-Flamingo	VLA	ABCD→D	**0.964**	0.896	0.824	0.740	0.660	4.09
GR-1	Prediction	ABCD→D	0.949	0.896	0.844	0.789	0.731	4.21
UP-VLA	Prediction&VLA	ABCD→D	0.962	**0.921**	**0.879**	**0.842**	**0.812**	**4.42**

Table 2. Zero-shot long-horizon evaluation on the Calvin ABCD→D benchmark. Results of baselines are copied from original papers.

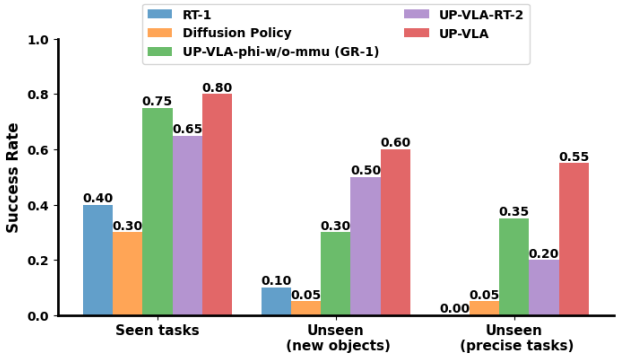

Figure 6. Real-world Panda robot manipulation results. UP-VLA achieves 80% success on seen-object tasks and 58% on unseen-object tasks. The strong unseen-object performance reflects the semantic generalization enabled by MMU pre-training.

5.4. Real-World Manipulation Results

Figure 6 shows real-world results. UP-VLA achieves **80%** success on seen-object tasks and **58%** on unseen-object tasks. The ablation results (Table 3) show that without MMU training, unseen-object performance collapses to only 20%, underscoring that semantic understanding from VQA pre-training is the primary driver of object-level generalization. The prediction component contributes complementary spatial precision, as seen from the 35% performance without prediction.

5.5. Ablation Study

Method	ABC→D	Real World	
	Avg.Len↑	Seen↑	Unseen↑
w/o MMU	3.89	0.85	0.20
w/o Bridge-Pretrain	2.74	0.65	0.30
w/o Prediction	1.44	0.65	0.35
w/o MMU-Condition	3.99	0.80	0.50
Full	4.08	0.80	0.58

Table 3. Ablating components of UP-VLA.

Table 3 systematically ablates each component of UP-VLA:

Removing future prediction (w/o Prediction) causes the most severe degradation on CALVIN ABC→D: the average completion length drops from 4.08 to 1.44, a collapse of nearly three full tasks. This strongly confirms that future visual prediction is essential for capturing low-level spatial dynamics that purely semantic representations cannot provide.

Removing MMU (w/o MMU) has a moderate effect on simulation performance (4.08→3.89) but catastrophically

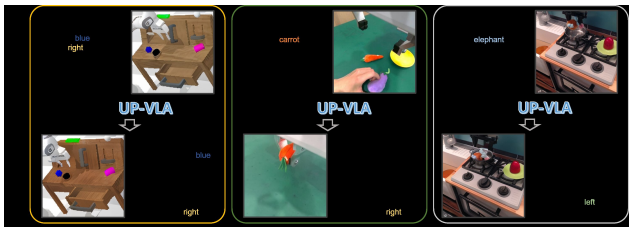

Figure 7. Qualitative results. Top: VQA outputs from UP-VLA demonstrating accurate scene understanding (object colors, positions, task state). Bottom: Predicted future images showing physically plausible object motion and gripper trajectories.

hurts real-world unseen-object generalization (0.58→0.20). This reveals that language-grounded semantic understanding is the key factor enabling the policy to transfer to novel objects and scenes.

Removing Bridge pre-training (w/o Bridge-Pretrain) reduces simulation performance to 2.74, highlighting that domain-relevant video pre-training for the PRE objective is crucial. The model cannot adequately learn robot-specific spatial dynamics from image-text data alone.

Removing MMU conditioning (w/o MMU-Condition) during action prediction results in smaller but consistent drops on both simulation (4.08→3.99) and real-world unseen (0.58→0.50), confirming that self-generated scene descriptions provide useful semantic context for action generation.

Notably, the interaction between MMU and PRE reveals a synergistic dynamic beyond simple additive gains. When both components are present, the MMU-generated scene descriptions provide structured semantic context that guides the prediction module toward task-relevant regions of the visual field, reducing prediction errors in cluttered scenes. Conversely, the spatial representations learned through the PRE objective enrich the backbone's feature space in ways that improve MMU accuracy on spatially demanding queries such as relative position estimation. This bidirectional reinforcement explains why the full model's performance exceeds the sum of individual component contributions, particularly on long-horizon tasks where accumulated errors from either semantic misinterpretation or imprecise spatial prediction are compounded across sequential sub-tasks.

Together, the ablations validate the design choices of UP-VLA: both MMU and PRE are individually necessary, and the combination produces synergistic benefits that neither achieves alone.

5.6. Qualitative Analysis

Figure 7 illustrates UP-VLA's dual capabilities. The top rows show accurate VQA outputs describing manipulation scenes—UP-VLA correctly identifies object attributes, spatial relationships, and task progress. The bottom rows show predicted future images: UP-VLA learns to anticipate object displacements, approach trajectories, and contact events with high visual fidelity. These physically coherent predictions serve as an implicit world model, enabling the action module to plan with greater precision and robustness.

Interestingly, the quality of predicted future frames correlates with action performance across task categories. Tasks requiring precise contact—such as pushing a button or rotating a switch—are associated with sharper, more geometrically consistent predictions, suggesting that the PRE objective implicitly learns fine-grained affordance representations beyond simple motion blur. In contrast, tasks with significant occlusion or deformable objects produce noisier predictions, which aligns with the observed drop in success rates for those task sub-categories. These observations motivate future work on uncertainty-aware prediction objectives that can signal when visual forecasts are unreliable and prompt the model to fall back on semantic reasoning.

6. Conclusion

We presented UP-VLA, a unified Vision-Language-Action model that co-optimizes multi-modal semantic understanding and future visual prediction within a single autoregressive framework. By employing dual visual streams—a CLIP-ViT encoder for semantic MMU and a VQ-GAN/MAGVITv2 tokenizer for discrete PRE—and routing both objectives through a shared Phi-1.5 backbone with a task-aware prompting and attention mechanism, UP-VLA acquires both high-level language grounding and low-level spatial dynamics awareness. A two-stage training procedure first co-pretrains on large-scale image-text data (LLaVA-665k) and robot video data (Bridge), then fine-tunes for action prediction with continued MMU co-training.

Experiments on the CALVIN benchmark demonstrate state-of-the-art performance: UP-VLA achieves an average completion length of 4.08 on the challenging ABC→D zero-shot generalization split (+33% over GR-1) and 4.42 on ABCD→D. On a real Franka Panda robot, UP-VLA attains 80% seen-object and 58% unseen-object success rates. Ablation studies confirm that both objectives are individually essential: prediction provides critical spatial dynamics information, MMU provides semantic generalization to novel objects, and their combination yields performance beyond what either achieves alone.

UP-VLA demonstrates that bridging semantic understanding with predictive world modeling is a highly effective strategy for embodied agents. We hope this work motivates further exploration of unified objectives for robot learn-

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ing, including extensions to 3D prediction, longer-horizon planning, and richer multimodal grounding signals. One particularly promising direction is uncertainty-aware visual prediction, which would allow the model to detect when its forecasts are unreliable and defer to semantic reasoning accordingly. Scaling UP-VLA's unified pre-training to larger and more diverse robot video collections is another natural avenue for substantially improving real-world robustness and cross-domain generalization.

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